

UNIT: 02 – AIR AND POLLUTION



Fig. Air pollution

Air pollution:

- Air pollution happens when harmful substances get into the air, making it unhealthy to breathe and damaging the environment.
- These harmful substances are called pollutants which come from various sources, such as:
 - (i) **Factories and power plants:** Emit smoke and chemicals.
 - (ii) **Vehicles:** Cars, trucks, and buses release exhaust fumes.
 - (iii) **Farming:** Produces gases like ammonia and methane.
 - (iv) **Homes:** Burning wood or coal for heating and cooking releases smoke.
 - (v) **Nature:** Forest fires, volcanoes and dust storms add pollutants to the air.
- These pollutants can cause health problems like breathing difficulties and heart disease, harm plants and animals and contribute to climate change.
- Reducing air pollution involves cutting down emissions from factories, vehicles, and farms and using cleaner energy sources.

Natural sources of air pollution:

Following are the natural sources of air pollution:

- (i) **Volcanoes:** Release gases and ash.
- (ii) **Forest Fires:** Produce smoke and gases.
- (iii) **Dust Storms:** Spread dust particles.
- (iv) **Pollen:** Release tiny particles into the air.
- (v) **Oceans:** Emit salt and organic compounds.
- (vi) **Soil Erosion:** Creates dust.
- (vii) **Animals:** Release gases like methane.
- (viii) **Plants and Trees:** Release organic compounds.
- (ix) **Lightning:** Produces nitrogen gases.
- (x) **Radon:** Naturally occurring but contributes minimally to air pollution



Fig. Natural Sources of Air Pollution

Man Made sources of air pollution:

- (i) **Factories:** Release smoke and gases.
- (ii) **Cars and Trucks:** Emit exhaust fumes.
- (iii) **Home Heating:** Burning coal, oil, and gas.
- (iv) **Farming:** Use of fertilizers and animal waste.
- (v) **Construction:** Produces dust.
- (vi) **Waste Burning:** Releases harmful gases and particles.
- (vii) **Cleaning Products:** Release chemicals into the air.
- (viii) **Mining:** Produces dust and gases.
- (ix) **Power Plants:** Burn fossil fuels, releasing smoke and gases.
- (x) **Planes and Ships:** Emit exhaust and smoke.



Fig. Man-made Sources of Air Pollution

Air pollutants and Types:

- Air pollutants are harmful substances in the air that can affect human health, the environment and the climate.
- Here are the main types of air pollutants:

(i) Particulate Matter (PM)

- Tiny particles like dust and smoke.

(ii) Carbon Monoxide (CO)

- A colorless, odorless gas from burning fossil fuels (like car exhaust).

(iii) Sulfur Dioxide (SO₂)

- A gas from burning fossil fuels with sulfur (like from power plants).

(iv) Nitrogen Oxides (NO_x)

- Gases from burning fuels (like from cars and power plants).

(v) Ozone (O₃)

- A harmful gas at ground level formed by sunlight reacting with other pollutants, while ozone in the upper atmosphere protects us from UV radiation.

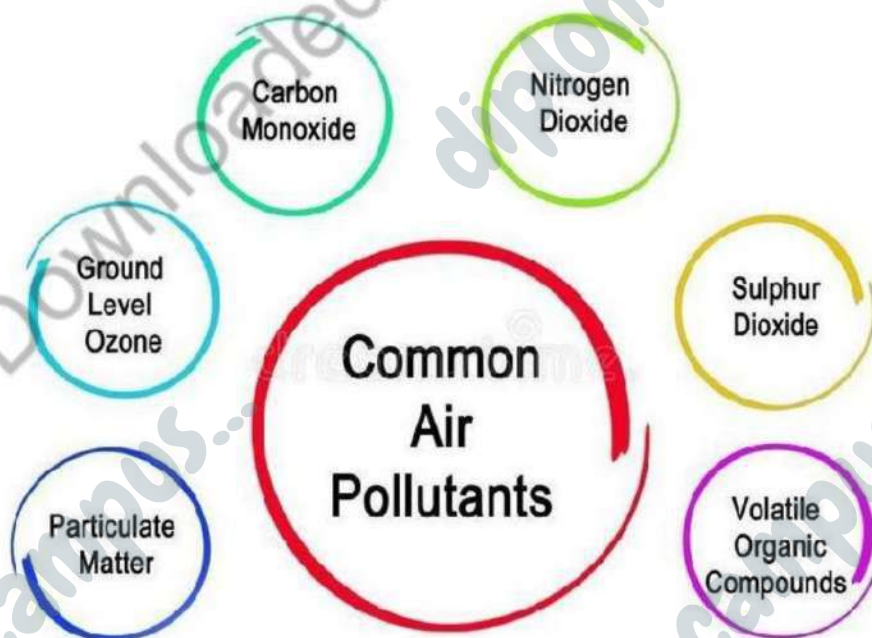


Fig. Air Pollutants

Particulate Pollutants and their effects:

- Particulate pollutants are tiny solid particles or liquid droplets suspended in the air.
- They come from various sources, both natural and human-made and can vary in size and composition.
- Common sources include:
 - (i) **Combustion:** From vehicles, power plants, forest fires and burning of biomass or fossil fuels.
 - (ii) **Industrial Processes:** Such as manufacturing, construction and mining activities.
 - (iii) **Natural Sources:** Like dust storms, volcanic eruptions and sea spray.
- These particles can range in size from very small (PM_{2.5}, PM₁₀) to larger particles visible as smoke or dust.
- They can have significant impacts on human health, air quality, visibility and even contribute to climate change.

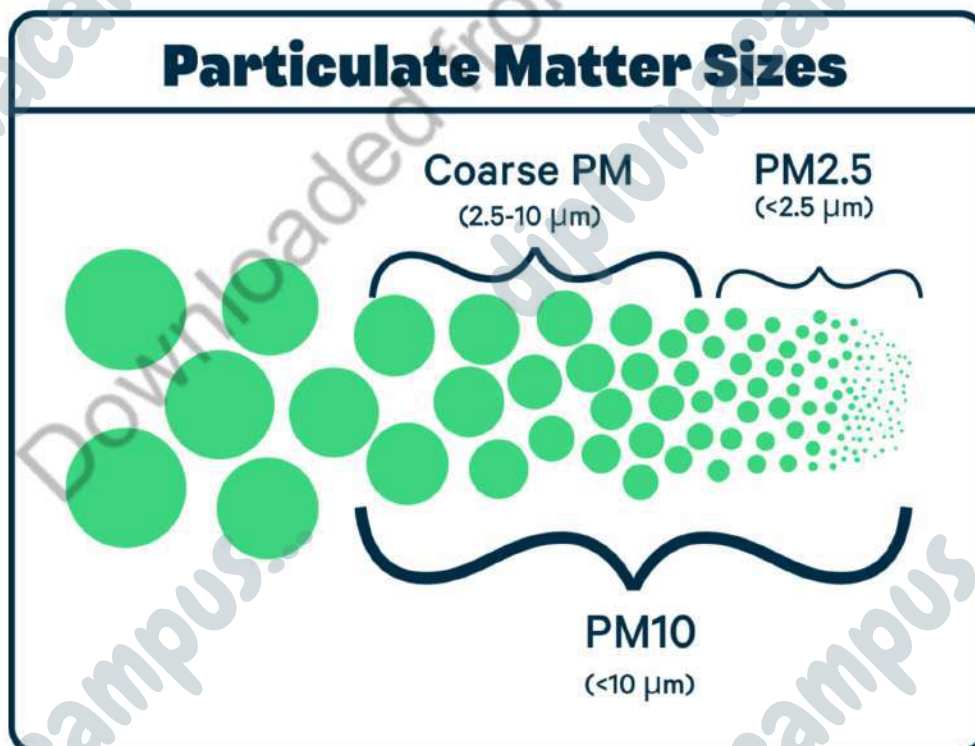


Fig. Size of Particulate Matter (PM) in micrometre (µm)

Effects of Particulate Pollutants:

- (i) **Health Problems:** Can cause breathing issues, like asthma etc.
- (ii) **Heart Issues:** Increase risks of heart attacks and strokes.
- (iii) **Shorter Lives:** Long-term exposure may lead to premature death.
- (iv) **Bad Air:** Contribute to smog and make it hard to see.
- (v) **Climate Change:** Some types can warm the Earth by trapping heat.



Fig. Effects of Particulate Pollutants (poster)

Control of Particulate Pollutants by Cyclone Separator:

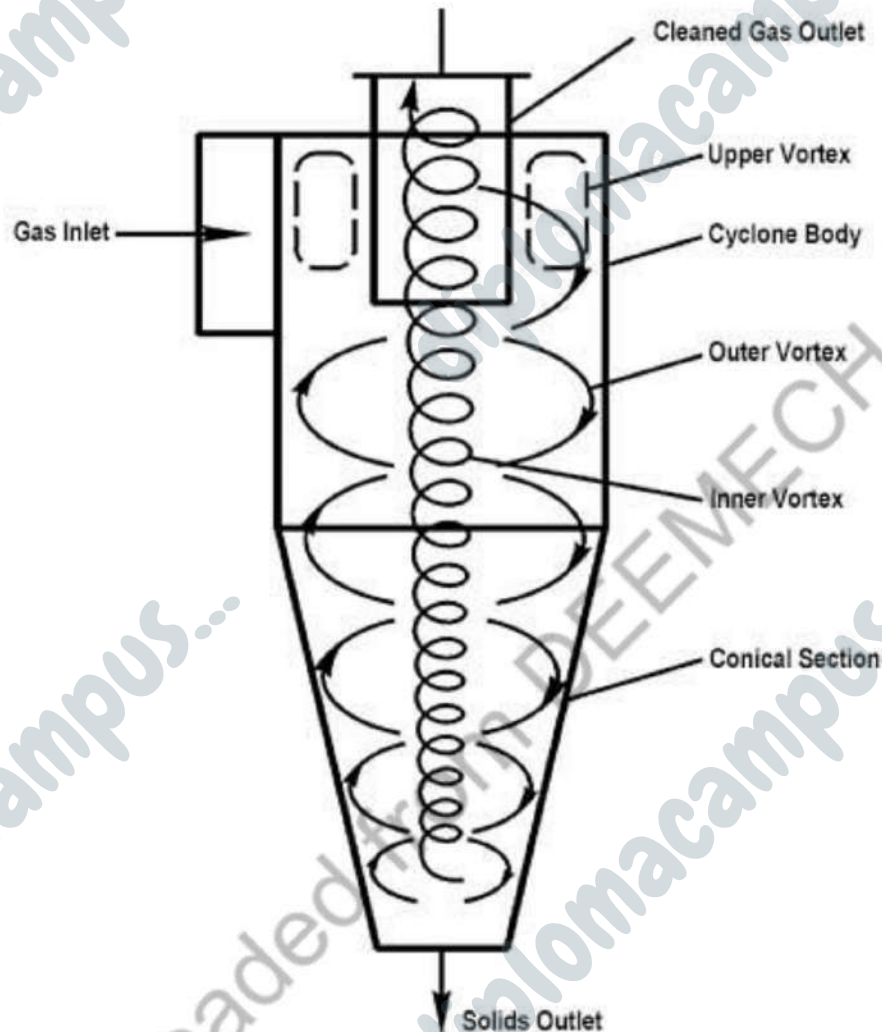


Fig. Cyclone Separator

- A cyclone separator is a device used to remove particulate matter from an air, gas or liquid stream without the use of filters, through vortex separation.

How Cyclone Separators Work:

- (i) **Gas Inlet:** Polluted air enters the cyclone separator at high speed, tangentially, which creates a spiral or vortex motion.
- (ii) **Vortex Motion:** The circular motion forces heavier particles toward the outer walls due to centrifugal force.

- (iii) **Downward Spiral:** The particles then move down the cyclone wall and collect in a collection bin at the bottom (solid outlet).
- (iv) **Cleaned gas outlet:** The cleaned air spirals up through the centre of the cyclone and exits through an outlet at the top.

Advantages of Cyclone Separators:

- Effective at removing large and medium-sized particles.
- Low Maintenance no moving parts, make it maintenance-free.
- Lower initial and operational costs compared to other filtration systems.
- Durable, can handle high temperatures and abrasive materials.

Limitations of Cyclone Separators:

- Not Suitable for Very Fine Particles
- Can cause a significant pressure drop in the system, requiring additional energy for air movement.

Applications of Cyclone Separators:

- **Industrial Processes:** Used in industries such as cement, steel, wood and chemical manufacturing.
- **Air Pollution Control:** Installed in air pollution control systems to reduce particulate emissions.
- **Dust Collection:** Employed in dust collection systems in woodworking shops and other facilities.

Control of Particulate Pollutants by Electrostatic Precipitator:

- An electrostatic precipitator (ESP) is a device used to remove fine particles from a flowing gas using the force of an induced electrostatic charge.
- ESPs are widely used in various industries to reduce air pollution by removing particulates from exhaust gases before they are released into the atmosphere.

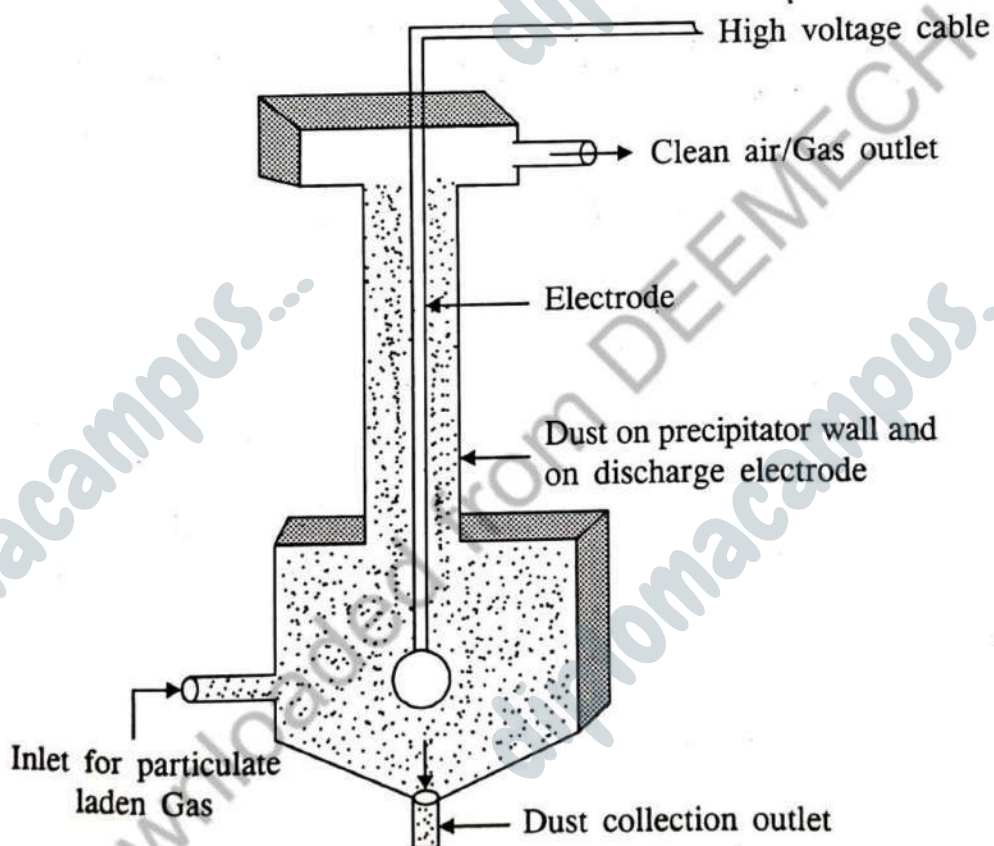


Fig. Electrostatic Precipitator (ESP)

Working Principle:

- The gas stream containing particles enters the ESP.
- High-voltage electrodes (discharge electrodes) create a strong electric field.
- As the gas and particles pass through the electric field, the particles become charged (usually negatively).

- The charged particles are drawn towards the collecting plates, which are having an opposite charge.
- The clean gas, now free of most particulates, exits the ESP and is either released into the atmosphere or further processed.

Advantages of Electrostatic Precipitator (ESP):

- High efficiency in removing fine particles.
- Can handle large volumes of gas.
- Low operating costs once installed.
- Can be used for both small and large particulates

Disadvantages of Electrostatic Precipitator (ESP):

- High initial cost.
- Requires regular maintenance and periodic cleaning.
- Not effective for gaseous pollutants.

Applications of Electrostatic Precipitator (ESP):

- Power plants (to remove fly ash from flue gas)
- Cement factories (to remove cement dust)
- Pulp and paper mills
- Steel plants (to remove metallic dust)
- Chemical plants

Air (prevention and control of pollution) act 1981.

- The Air (Prevention and Control of Pollution) Act, 1981, is a law in India aimed at controlling and reducing air pollution.



i. Purpose:

- To prevent and control air pollution.
- The Act helps India manage and reduce air pollution to protect the environment and public health.

ii. Boards:

- Central Pollution Control Board (CPCB) and State Pollution Control Boards (SPCBs) are set up to enforce the law.

iii. Functions of the Boards:

- Help the government with air pollution issues.
- Setting standards for safe air quality levels and emission limits for industries and vehicles.
- Check air pollution levels and inspect pollution sources.
- Enforce Rules

iv. Regulations:

- Industries need permission from SPCBs to operate.
- Controls are placed on vehicle emissions and industrial pollution.

v. Penalties:

- Violating the rules can lead to fines and jail time.

vi. Public Involvement:

- Encourages people to participate in pollution control through awareness and public hearings.

Questions on Unit: 02 – Air and Pollution (Cognitive Level: R & U)

1. What is air pollution and list the sources of pollutants.
2. List the natural source of air pollution.
3. List the man-made source of air pollution.
4. What are air pollutants and list the types of air pollutants.
5. Explain Particulate pollutants.
6. What are the effects of Particulate pollutants?
7. Explain the working of cyclone separator with the help of neat diagram.
8. List the advantages, disadvantages and applications of cyclone separator.
9. Explain the working of electrostatic precipitator with the help of neat diagram.
8. List the advantages, disadvantages and applications of electrostatic precipitator.
9. What are the purpose and functions of Central Pollution Control Board (CPCB) and State Pollution Control Boards (SPCBs).
10. Explain Air (prevention and control of pollution) act 1981.

End of Unit: 02

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